

Activity: regression with multiple explanatory variables

Group members:

Interactions

Here is the equation for a fitted model on the `penguins` data, with bill length as the response, and an interaction between body mass and species:

$$\widehat{\text{bill length}} = 26.99 + 5.18 \text{ IsChinstrap} - 0.26 \text{ IsGentoo} + 0.0032 \text{ body mass} \\ + 0.0013 \text{ body mass} \cdot \text{IsChinstrap} + 0.0010 \text{ body mass} \cdot \text{IsGentoo}$$

Questions

1. What is the estimated bill length of a Gentoo penguin with a body mass of 5000g?
2. What is the estimated slope on body mass, for Chinstrap penguins?